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Track 1

AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

Extend existing
Redrob
capabilities

Improve
experiences for
Redrob users

Introduce
new AI-native
workflows

Strengthen ecosystem
engagement and
network effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

How the idea leverages existing Redrob capabilities

What new capability, workflow, or value layer is introduced

How the idea strengthens the overall Redrob ecosystem

Why Redrob is uniquely positioned to enable this experience

EMMA

Enterprise Metacognitive Multi-Agent Fleet



Challenge 1: AI Systems Architect



42-Day Build



Reimagining Work

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Track 01

AI Systems & Workflow Innovation Challenge

Team Name : Metacognitive Fleet

Problem Statement: Securing Autonomous local AI Developers against Infinite Loops, Resources Exhaustion, and Context Drift under strict On-Premise constraints.

Team Members : Sourav Ray (Lead Systems & Cognitive Architect)

PROBLEM DEFINITION - 01

The Code Execution Hazards of Autonomous Agents

- **The Runway Code Loop:**

-> AI models often generate accidental infinite loops (like `while True: Pass`) or deep recursion. These lock up 100% of CPU capacity, freezing the environment and stalling development tasks.

- **Silent Out-of-Memory (OOM) Kills:**

-> Heavy mathematical operations and data processing can allocate memory uncontrollably. The OS silently terminates the process, leaving the agent blind with zero diagnostic logs to learn from.

- **Goal Drift & Infinite Iteration Stalls:**

-> Without strict logic boundary monitors, developers agents get trapped in endless “bug-fixing loops” – generating the same invalid code patterns repeatedly without detecting they have failed.



PROBLEM DEFINITION -02

THE TARGET PROFILES AND PERSONAS AFFECTED BY AGENTIC EXECUTION RISKS

- Enterprise Software Engineering Teams:

Need: Want to automate legacy database integration, but blocked by security concerns over letting AI run code directly on servers.

- Local AI Developers & Researchers:

Need: Individual developers running local coding assistants who require isolated sandboxes to protect the host machines from directory access.

- DevOps & Platform Engineers:

Need: Technical operators who must enforce strict resource boundaries (CPU limits, RAM caps) to prevent agents from crashing environments.



PROBLEM DEFINITION - 03

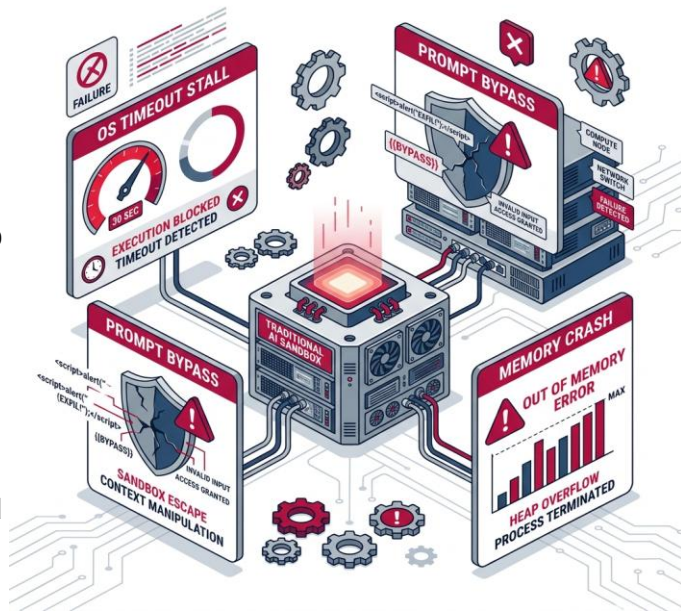
THE BOTTLENECKS OF STANDARD AGENT SANDBOXES

- **Slow OS-Level Timeouts:**

The Failure: Standard environments wait for a 30-second OS wall-clock timeout to kill a frozen process. This wastes valuable CPU capacity and stalls execution pipelines.

- **Fragile, Prompt-Level Rules:**

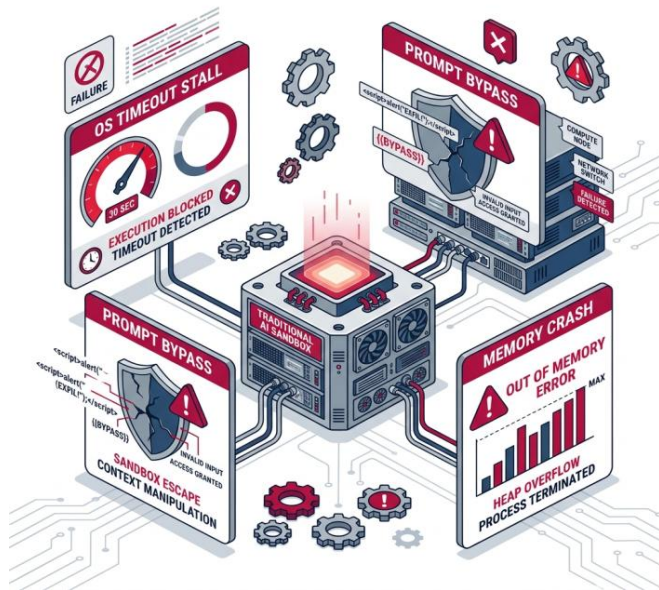
The Failure: Instructing a model to "be safe" via system prompts is easily bypassed by code errors or logical hallucinations. Soft prompts are not hard security boundaries.



PROBLEM DEFINITION - 03

- **Rigid, Non-Adaptive Resource Caps:**

The Failure: Fixed sandboxes apply static resource ceilings, Safe, complex algorithms (such as matrix transformations) fail simply because they cannot request scaled CPU or gas parameters dynamically.



- **No Causal Convergence or Checkpoints:**

The Failure: Standard executors lack transaction tracking. When an agent gets caught in a bug loop, it blindly overwrites files or crashes, with no automated git-backed rollback checkpoints.

Opportunity & Vision

Questions

- Why is this an important opportunity?
- What future state are you enabling?

Solution Overview

Questions

- What is your proposed solution?
- What makes it AI-native rather than AI-assisted?
- Which existing Redrob capability, workflow, or user journey does your solution build upon?

User Journey / Workflow Diagram

Questions

- How does a user interact with the solution?
- How does information flow through the process?
- Where does the solution integrate with Redrob experiences?

Mandatory Visual

- Workflow Diagram

AI Logic & Decision Flow

Questions

- Where does AI intervene?
- How are decisions made?
- How do agents/models/reasoning systems interact?

Mandatory Visual

- AI Flow Diagram

System Architecture

Questions

- What components make up the system?
- How do services interact?
- Which Redrob systems, workflows, or services are leveraged?

Mandatory Visual

- Architecture Diagram

Data, Context & Intelligence Layer

Questions

- What data powers the solution?
- How is context retrieved, stored, or utilized?
- How does existing Redrob context, signals, or workflows improve the experience?

Mandatory Visual

- Data Flow Diagram

Scalability & Technical Feasibility

Questions

- How would this be implemented?
- How does the system scale?
- What technical challenges exist?

Redrob Ecosystem Integration

Questions

- Which existing Redrob capabilities are being leveraged?
- What new capability does your solution introduce?
- How does it strengthen the overall Redrob ecosystem?
- What additional opportunities become possible after implementation?

Mandatory Visual

- Ecosystem Integration Diagram

Impact & Success Metrics

Questions

- What measurable outcomes are expected?
- How will success be tracked?
- What value is created for users and Redrob?

Future Roadmap

Questions

- How could this evolve over 2–3 years?
- What future capabilities can be unlocked?
- What broader vision does this support?



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THANK YOU